

3 Atomic Data

3.1 Ba

ω_K	:	0,900	(4)
$\bar{\omega}_L$	:	0,110	(5)
n_{KL}	:	0,888	(4)

3.1.1 X Radiations

	Energy (keV)		Relative probability
X _K			
Kα ₂	31,8174		54,28
Kα ₁	32,1939		100
Kβ ₃	36,3045	}	29,41
Kβ ₁	36,3786		
Kβ ₅ ''	36,654		
Kβ ₂	37,258	}	7,41
Kβ ₄	37,312		
KO _{2,3}	37,425		
X _L			
Lℓ	3,9544		
Lα	4,4515 - 4,4666		
Lη	4,3307		
Lβ	4,8278 - 5,207		
Lγ	5,3715 - 5,8104		

3.1.2 Auger Electrons

	Energy (keV)	Relative probability
Auger K		
KLL	25,314 - 26,786	100
KLX	30,095 - 32,179	47,7
KXY	34,86 - 37,41	5,7
Auger L		
	2,6614 - 5,8064	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
e _{AL}	(Ba)	2,6614 - 5,8064	7,39 (5)
e _{AK}	(Ba)		
	KLL	25,314 - 26,786	} 0,78 (4)
	KLX	30,095 - 32,179	
	KXY	34,86 - 37,41	
ec _{2,0 T}	(Ba)	624,216 - 661,642	9,56 (14)
ec _{2,0 K}	(Ba)	624,216 (3)	7,78 (11)
ec _{2,0 L}	(Ba)	655,668 - 656,410	1,401 (20)
ec _{2,0 M}	(Ba)	660,364 - 660,876	0,2992 (43)
$\beta_{0,2}^-$	max:	513,97 (17)	} 94,57 (26)
	avg:	173,67 (6)	
$\beta_{0,1}^-$	max:	892,17 (18)	} 0,0006 (1)
	avg:	332,51 (7)	
$\beta_{0,0}^-$	max:	1175,63 (17)	} 5,43 (26)
	avg:	284,90 (5)	

5 Photon Emissions

5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)	
XL	(Ba)	3,9544 - 5,8104	0,919 (16)	
XK α_2	(Ba)	31,8174	1,99 (4)	} K α
XK α_1	(Ba)	32,1939	3,66 (6)	
XK β_3	(Ba)	36,3045	} 1,078 (20)	K' β_1
XK β_1	(Ba)	36,3786		
XK β_5''	(Ba)	36,654		
XK β_2	(Ba)	37,258	} 0,272 (8)	K' β_2
XK β_4	(Ba)	37,312		
XK $\alpha_{2,3}$	(Ba)	37,425		

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{1,0}(\text{Ba})$	283,46 (7)	0,0006 (1)
$\gamma_{2,1}(\text{Ba})$	378,20 (7)	0,0000106 (9)
$\gamma_{2,0}(\text{Ba})$	661,655 (3)	85,01 (20)

6 Main Production Modes

The ¹³⁷Cs is mainly produced by thermal neutron-induced fission of the ²³⁵U in nuclear reactors.

7 References

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